Experiment 10: Text Summarization

A. Gensim with TextRank

```
!pip -q install sumy
text = """
Natural Language Processing is a branch of artificial intelligence.
It enables computers to understand and process human language.
NLP is used in machine translation, sentiment analysis and chatbots.
Text summarization is an important application of NLP.
Extractive summarization selects important sentences from the original text.
Abstractive summarization generates new sentences that convey the main ideas
TextRank ranks sentences based on their relationships and importance.
"""

summary = ""

# Try the original Gensim TextRank summarizer
try:
    from gensim.summarization import summarize

    summary = summarize(text, ratio=0.4)

    if not summary.strip():
        raise ValueError("Gensim returned an empty summary.")

    method = "Gensim TextRank"

except (ImportError, ModuleNotFoundError, ValueError, AttributeError):
    # Compatible TextRank fallback for modern environments
    from sumy.parsers.plaintext import PlaintextParser
    from sumy.nlp.tokenizers import Tokenizer
    from sumy.summarizers.text_rank import TextRankSummarizer
    from sumy.utils import get_stop_words

    parser = PlaintextParser.from_string(
        text, Tokenizer("english")
    )

    summarizer = TextRankSummarizer()
    summarizer.stop_words = get_stop_words("english")

    sentences = list(summarizer(parser.document, 3))
    summary = " ".join(str(sentence) for sentence in sentences)
    method = "Sumy TextRank compatibility fallback"

print("Summarization Method:", method)
```

```
    print("\nInput Text:\n", text)
    print("\nSummary:\n", summary)
```

Next steps: Explain error

```
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
  Building wheel for docopt (pyproject.toml) ... done
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
/tmp/ipykernel_6231/2258972738.py in <cell line: 0>()
     15 try:
---> 16     from gensim.summarization import summarize
     17

ModuleNotFoundError: No module named 'gensim'

During handling of the above exception, another exception occurred:

LookupError                               Traceback (most recent call last)
                                          6 frames
LookupError:
**********************************************************************
  Resource punkt_tab not found.
  Please use the NLTK Downloader to obtain the resource:

  >>> import nltk
  >>> nltk.download('punkt_tab')

  For more information see: https://www.nltk.org/data.html

  Attempted to load tokenizers/punkt_tab/english/

  Searched in:
    - '/root/nltk_data'
    - '/usr/nltk_data'
    - '/usr/share/nltk_data'
    - '/usr/lib/nltk_data'
    - '/usr/share/nltk_data'
    - '/usr/local/share/nltk_data'
    - '/usr/lib/nltk_data'
```

B. Text summarization using Transformers (GPT-2)

```
!pip -q install transformers torch
from transformers import pipeline

text = """
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"""

# GPT-2 is a general text-generation model, not a summarization-finet
generator = pipeline(
    "text-generation",
    model="gpt2"
)
```